Technical Specifications

Item	Specification	Statement of Compliance
1	1. The system must have interfaces, enabling all NPC officials and employees to manage inventory for supplies to generate or process reports and forms. The following modules and functionalities must be integrated into the Inventory Management System (IMS).	
	1.1 **Dashboard** 1.1.1 Overview: Displays a summary of essential inventory metrics, such as total stock levels, value, and critical alerts. 1.1.2 Visual Representation: Presents inventory data using simple charts or graphs for quick insights, like bar charts for stock levels and pie charts for category distribution. 1.1.3 Alerts: Notifies users about low stock levels, or other important inventory events. 1.1.4 Historical Data: Allows users to view historical inventory data to identify trends and make informed decisions.	
	1.2 **Application Set-up and libraries Module** 1.2.1 Departments: Create and organize different departments. 1.2.2 Asset Categories/Classification: Define categories for ICT software, real assets, and vehicular assets, supplies and materials, semi-expendables, and property, plant, and equipment.	

	1.2.3 Asset Locations: Set up and manage locations where assets are stored. 1.2.4 Suppliers: Organize and categorize suppliers for efficient management.	
	1.3 **User Account Management Module –** IMS must be customizable to manage user accounts and permissions based on requirements of the end-user and recommendation of the NPC Interim Management Information System Unit (IMISU). Allows users and administrators to access and manage the system based on the roles. 1.3.1 Must have Single Sign-on (SSO) capability 1.3.2 User login 1.3.3 Audit trails and activity logs 1.3.4 Accounts and role assignments	
	1.4 **Inventory Equipment Management Module –** Manage all moveable/non moveable assets, properties, including but not limited to description, acquisition cost and date, accountable personnel, status and list of items assigned to personnel, division or unit. 1.4.1 Must be able to monitor all property transactions (Receiving, Issuance, Return, Transfer, Updating, Repair, Disposal). 1.4.2 Must be able to generate forms, and reports for the following transactions: 1.4.2.1 Gate Pass 1.4.2.2 Purchase Order or Contract related to the Procurement of Capital Outlays 1.4.2.3 Property Acknowledgment Receipt (PAR)	

		1.4.2.4 Inventory Custodian Slip (ICS) 1.4.2.5 Property Card (PC) 1.4.2.6 Report on the Physical Count of Property, Plant and Equipment (RPCPPE) 1.4.2.7 Property Inventory Form (PIF) 1.4.2.8 Inventory and Inspection Report of Unserviceable Properties (IIRUP) 1.4.2.9 Invoice Receipt for Property 1.4.2.10 Waste Material Report (WMR) 1.4.2.11 Report of Lost, Stolen, Damaged or Destroyed Property (RLSDDP) 1.4.2.12 Property Transfer Report (PTR) 1.4.2.13 Inspection and Acceptance Report (IAR) 1.4.2.14 Supplier Maintenance 1.4.2.15 Property Card 1.4.2.16 Property, plant and equipment ledger card 1.4.2.17 Inspection and Acceptance Report monitoring 1.4.2.18 Depreciation Lapsing Schedule 1.4.2.19 Certificate of Property turn over (for clearance) 1.4.3 Must be able to generate series/barcodes/codes/property number or tag. 1.4.4 Must be able to upload picture/image of each asset. 1.4.5 Must be able to set, compute and monitor assets that are for maintenance and disposal. 1.4.6 Must be capable of evaluation and recognition of assets and its inventory. 1.4.7 Must be capable of monitoring the location and whose accountability.	

	1.4.8 Must be capable of showing the history of issuances and transfer of asset accountabilities. 1.4.9 Must be ready for each user's viewing of all assigned assets. 1.4.10 Must be able to change or edit signatories of reports. 1.4.11 Must be able to integrate with a digital signature platform via API with cryptographic key technology. 1.4.12 Must be able to monitor turn over and issuance of certificate for clearance. 1.4.13 Must be able to compute depreciation values and monitor lifespan of assets according to Government Accounting Manual (GAM). 1.4.14 Must be able to assign, scan and read barcodes/QR codes during annual inventory and update details of each asset real-time.	
	1.5 **Inventory Supplies Management Module -** Manage and monitor all common-used supplies for issuance, consumption and costs. 1.5.1 Must be able to monitor all common-used supplies transactions (Receiving, Issuance, Return, Transfer, Updating, Repair as applicable and Disposal) 1.5.2 Must be able to generate forms and reports such as: 1.5.2.1 Certificate of non-availability of stock (CNAS) 1.5.2.2 Purchase Order related to the Procurement of CSE Supplies 1.5.2.3 Inventory Registry of semi-expendable property issued 1.5.2.4 Inspection and Acceptance Report (IAR)	

		1.5.2.5 Inventory Transfer Report (ITR) 1.5.2.6 Letter of Undertaking for Contact of Service (COS) personnel 1.5.2.7 Requisition and Issue Slip (RIS) 1.5.2.8 RIS Monitoring 1.5.2.9 Inventory Custodian Slip (ICS) 1.5.2.10 Semi-expendable Property Ledger Card 1.5.2.11 Stock Card (SC) 1.5.2.12 Supply Ledger Card 1.5.2.13 Report on Supplies and Materials Issued (RSMI) 1.5.2.14 Report on semi-expendable property issued 1.5.2.15 Report on the Physical Count of Inventories (RPCI) (for supplies) 1.5.2.16 Report on the Physical Count of Semi-expendable Property (for semi-expendable and fixture, semi-expendable office equipment – ICT and semi-expendable office equipment) 1.5.2.17 Receipt of returned semi-expendable property 1.5.2.18 Inventory and Acceptance Report monitoring 1.5.2.19 Waste Material Report (WMR) 1.5.2.20 Report of Lost, Stolen, Damaged or destroyed semi-expendable Property 1.5.2.21 Inventory and inspection of unserviceable semi-expendable properties 1.5.2.22 List of available and commonly used supplies	

	1.5.3 Must be able to generate a series/barcodes/QR codes/property number or tag. 1.5.4 Must be able to upload picture/image of each asset. 1.5.5 Must be able to monitor turn over and issuance of certificate for clearance.	
	1.6 **Report and barcode/QR code generation** 1.6.1 Provide comprehensive insights into the status, utilization, valuation, and history of supplies, assets, depreciation value, inventory, enabling informed decision-making and efficient resource management within an organization. 1.6.2 Must be able to generate reports in CSV, Excel, PDF etc. for portability. 1.6.3 Must be able to generate unique barcode/QR for each asset and print individually or by series/batch. 1.6.4 Must be able to generate the following reports monthly, annually or by any given dates: 1.6.4.1 Asset Master list 1.6.4.2 Supplies/Item Master list 1.6.4.3 Supplier Master list 1.6.4.4 User Master list 1.6.4.5 Purchase Order (PO) 1.6.4.6 Generation of Summary of Purchase Order 1.6.4.7 Property Acknowledgment Receipt (PAR) 1.6.4.8 Property Card (PC) 1.6.4.9 Report on the Physical Count of Property, Plant and Equipment (RPCPPE) 1.6.4.10 Inventory and Inspection Report (IAI) 1.6.4.11 Waste Material Report (WMR)	

	1.6.4.12 Report of Lost, Stolen, Damaged or Destroyed Property (RLSDDP) 1.6.4.13 Request for Transfer of Accountability (RTA) 1.6.4.14 Gate Pass 1.6.4.15 Inspection and Acceptance Report (IAR) 1.6.4.16 Requisition and Issue Slip (RIS) 1.6.4.17 Inventory Custodian Slip (ICS) 1.6.4.18 Stock Card (SC) 1.6.4.19 Report on the Physical Count of Inventories (RPCI) 1.6.4.20 Reordering Point	
	2. The IMS must be capable of the initial build-up of a new IMS database and the migration of existing inventory management database to the new database. In this respect, it requires the following services:	
	2.1 The sizing and configuring of the necessary systems storage space and the set-up and establishment of the required database. 2.2 The provision of the necessary knowledge transfer that covers the IMS usage, including but not limited to, database management, administration, monitoring and maintenance. 2.3 The provision of 12 months warranty commencing from the issuance of the Certificate of User Acceptance as mentioned under Section XVI hereof. 2.4 Deployment Services: 2.4.1 **Cloud** The SUPPLIER shall assist in deployment services of the project ready for production using NPC cloud environment. These services	

		shall include the following but not limited to: 2.4.1.1 Preparation of the customized IMS to deploy on a cloud environment provided by the IMISU. 2.4.1.2 Application and configuration of Transport Layer Security (TLS) for data encryption that includes the deployment of certificate. 2.4.1.3 Configuration of instance 2.4.1.4 Domain and web server deployment and configuration	
		2.4.2 **On-Premises** The IMS must be capable for on-premises deployment. The SUPPLIER shall collaborate with IMISU to initiate the following acts, including, but not limited to the following: 2.4.2.1 Preparation of the application system to deploy on an on-premises environment specified by the end-user 2.4.2.2 Domain and web server deployment and configuration 2.4.2.3 Configuration of Virtual machines or remote desktop 2.4.2.4 Configuration and application of firewall 2.4.2.5 Supply and deployment of a plug-and-play hardware security device with honeypot, vulnerability assessment, and unlimited health checks and automated scans to enhance IMS on-premise protection.	

		2.4.2.6 The hardware security device should have received awards from Malaysia Cyber Security Awards on Cyber Security Innovation for at least two (2) consecutive years. 2.4.2.7 Implementation of embedded honeypot technology for early threat detection within the IMS network environment. 2.4.2.8 Enablement of proactive vulnerability assessment and continuous risk discovery for IMS services. 2.4.2.9 Provision of unlimited health checks, security scans, and reporting for IMS network infrastructure.	
	2.5 Quality, functionality and performance testing The SUPPLIER shall ensure that the IMS is tested accordingly based on the quality, functionality and performance following the end user's requirements. The SUPPLIER shall provide test plans and test case design and its results to the end user and IMISU. The types of testing that will be provided shall be the following, but not limited to: 2.5.1 The SUPPLIER shall provide the complete lifecycle procedures, test plans and test case designs for the approval of the end-user with the recommendation of the IMISU. The testing requirements for the Testing shall focus on the quality, functionality, and performance of the IMS. Other testing methods (e.g., regression testing, automated testing, performance testing etc.,) may be included or added to the	

	testing plan based on the end-user's decisions.	
	2.6 Security Testing 2.6.1 The SUPPLIER shall assist in third party Vulnerability Assessment and Penetration Tests (VAPT) provided by NPC.	
	2.7 Privacy Impact Assessment The SUPPLIER shall fully participate to the end user's conduct of privacy impact assessment (PIA) to the proposed application system. In the conduct of the said activity, the supplier shall provide the following: 2.7.1 Provide solutions to the identified privacy risks if applicable. 2.7.2 Assist in the determination of appropriate security measures identified during the conduct of PIA. 2.7.3 Submit documents pertaining to technical security measures and data protection initiatives as required during the conduct of PIA.	
	SUPPLIER CRITERIA/REQUIREMENTS 1. The Supplier must have on going or completed ICT related project with any government agency within the last 3 years with contract price of at least equal to herein ABC or higher. 2. The Supplier must have local support available from Monday to Friday except holidays from 8am to 5pm with the option to upgrade to 24x7 support. 3. The Supplier must not have a history of blacklisting (or any recommendations towards such) in PhilGEPS as raised by any government agency in the past five (5) years.	

	4. The Supplier has an updated National Privacy Commission (NPC) Data Processing System Registration and has a Data Protection Officer.	
	TECHNICAL REQUIREMENTS 1. The IMS must be web based and should be compatible with known web browsers. 2. The NPC has the option to request for two (2) revisions of the IMS design prior the acceptance and production deployment of the system. 3. The revision of IMS will be acknowledged by the supplier during the scheduled wireframe presentation.	
	OTHER FUNCTIONAL REQUIREMENTS Aside from the above-mentioned functionality, the IMS system should have the following features: 1. The IMS solution must be ready to provide workflow process that enable integration (via API) with the platform of digital signatures based on cryptographic key technology. 2. The platform must be ready to support the signing of documents using digital certificates issued by a Trusted Certificate Authority (CA) recognized by the Philippine Government, such as the PNPKI or NPC Private CA (if any).	
	DOCUMENTATION The following documentation are required to be submitted: 1. Technical documents: Installation and Configuration manual, operations manual, user's manual, API references (if applicable), technical specifications of the IMS; and 2. All other documents as may be necessary.	
	KNOWLEDGE TRANSFER A knowledge transfer plan shall be submitted which includes face to face method of delivery, specific topics to be covered and the corresponding date and times for each topic. The knowledge transfer shall be provided to end users as well as the IMISU. At a minimum, the knowledge transfer must include:	

	1. Discussion of the features of the IMS including but not limited to adding users, report generation, importing files; 2. Discussion of the system components of the IMS including but not limited to system configuration, installation; 3. How to perform incident response relating to the IMS; and 4. Other topics as deemed necessary by the end user Proof of completion of the required training to be conducted shall be the issuance of training certificates to the participants. The supplier must be available for consultations as may be necessary.	
	SERVICE LEVEL AGREEMENT Technical support must be provided either on-site, via telephone or email to resolve technical and other related problems based on a Service Level Agreement (SLA) with provisions for liquidated damages for non-compliance. Resolution must be delivered based on the following levels of severity: 1. Severity 1 Critical: Major system or component failure with critical impact on business process. Service has stopped. The problem must be resolved through immediate support (within 4 hours). This includes successful attacks to the system as well as hacking attempts. 2. Severity 2 High: Minor system or component failure with high impact on business process. The service is running but performance has degraded. The problem must be resolved immediately (within 1 working day). 3. Severity 3 Moderate: Operational use exhibits signs of non-compliance with agreed functionality. The service is running but certain functions do not work as indicated. The problem must be resolved at the soonest time possible (within 5 working days).	

4. Severity 4 Low: Service is running. Efficiencies would be achieved through desired improvements in the system components. Feature must be available in the next update.

V. PAYMENT AND DELIVERY

Payment shall be made upon Issuance of the Inspection and Acceptance Certificate by the NPC Inspection and Acceptance Committee upon recommendation by the end-user, but consistent with the following:

% of Contract Cost	Deliverable and Indicative Delivery Date
20 %	Upon the delivery and submission of project workplan within thirty (30) days after the issuance of the Notice to Proceed (NTP)
70 %	100% IMS deployed with the required customization and testing - Evaluation of requirements completed accompanied by Certificate of User Acceptance signed by end-user and completion of the conduct of Privacy Impact Assessment. The supplier has conducted the necessary customization.
10 %	Submission of technical documents including but not limited to source code, software documentation, user manual and conduct of knowledge transfer to the end-users as to the system's management, administration, monitoring and maintenance.
100 %	TOTAL

Payment shall be processed in accordance to the above schedule of payment and deliverables, and upon issuance of Inspection and Acceptance Report by the NPC

	Inspection and Acceptance Committee and Certificate of Acceptance by the End-User.	
	POST QUALIFICATION REQUIREMENT Aside from the post-qualification documents required to be submitted as specified in the Bidding Documents, the bidder should also submit Registration Certificate issued by the National Privacy Commission upon registration of the company's Data Protection Officer.	
	OWNERSHIP OF DATA All software, data, source code (completed customization of IMS) and resources pertaining to the system shall be owned exclusively by the NPC.	
	RETENTION, DELETION AND DISPOSAL OF DATA If applicable, all personal data must be retained only for as long as necessary for the fulfillment of the purposes for which data was obtained. For this purpose, the personal data processed by the SUPPLIER shall be disposed of after the fulfillment of its contract. Upon the fulfillment of the contract, any personal data processed by the SUPPLIER must be deleted and disposed of in accordance with the provisions of the Data Privacy Act, its Implementing Rules and Regulations, and other issuances of the NPC. The deletion and disposal of data shall be under the supervision of the NPC. After the deletion and disposal of the data, the SUPPLIER must issue a certification that the data were indeed deleted and disposed of properly.	
	WARRANTIES OF THE SUPPLIER The supplier warrants that it shall conform strictly with the terms and conditions of these Terms of Reference. The warranty period shall commence upon issuance of the Certificate of User Acceptance with the applicable period, which shall be for twelve (12) months to cover software upgrades, security patches and technical support. The supplier shall	

	coordinate with the end-user and the IMISU in the performance of their jobs.	
	CONFIDENTIALITY OF DATA The supplier shall document detailed procedures and techniques in identifying systems security risks and breaches and how such shall be handled. The supplier and all project staff shall be required to sign a non-disclosure agreement immediately at the start of the project. All systems to which the project staff of the supplier shall be granted access to, its components, parts, specifications, data, ideas, technology, and technical and non-technical materials (collectively referred to here as "Proprietary Information") are confidential and proprietary to National Privacy Commission. The supplier agrees to hold the Proprietary Information in strict confidence and further agrees not to reproduce, transcribe, or disclose the Proprietary Information to third parties without the prior written approval of the National Privacy Commission.	